

Vaibhav KRISHAN

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RESEARCH POSITIONS

PDF (TCS)	The Institute of Mathematical Sciences	July 2024-Present
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EDUCATION

Ph.D. COMPUTER SCIENCE AND ENGINEERING	Indian Institute of Technology Bombay Advised by Prof. Sundar Vishwanathan and Prof. Nutan Limaye	July 2017-May 2024
B.Tech. COMPUTER SCIENCE AND ENGINEERING	Indian Institute of Technology Bombay CPI: 7.59/10	July 2009-May 2013

JOURNAL PUBLICATIONS

Algorithmica	A #SAT Algorithm for Small Constant-depth Circuits with PTF gates with S. Bajpai, D. Kush, N. Limaye and S. Srinivasan Algorithmica 84, 1132-1162 (2022).
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CONFERENCE PUBLICATIONS

MFCS 2026	On CC^0 Lower Bounds for AND via Torus Polynomials with J. Sarma The 51st International Symposium on Mathematical Foundations of Computer Science, 2026 (to appear).
SAT 2026	On Proof Systems for #QBF with S. Chede, L. Chew and A. Shukla The 29th International Conference on Theory and Applications of Satisfiability Testing, 2026 (to appear).
ITCS 2026	Lower Bounds and Separations for Torus Polynomials with S. Vishwanathan The 17 th Innovations in Theoretical Computer Science Conference, 2026.
CSR 2021	Upper Bound for Torus Polynomials The 16th International Computer Science Symposium in Russia, 2021.
ITCS 2019	A #SAT Algorithm for Small Constant-depth Circuits with PTF gates with S. Bajpai, D. Kush, N. Limaye and S. Srinivasan The 10 th Innovations in Theoretical Computer Science Conference, 2019.

PREPRINTS

ECCC

Isolation Lemma for Directed Reachability and NL vs. L
with N. Limaye

TALKS AND PRESENTATIONS

Presentation

Lower Bounds and Separations for Torus Polynomials
ITCS 2026.

Video

Lower Bounds and Separations for Torus Polynomials
ITCS 2026.

Lecture

Polynomials and Computation
Foundational Lecture Series on TCS, IMSc, 2025.

Presentation

Upper Bound for Torus Polynomials
CSR 2021. (online)

Presentation
and **Poster**

A #SAT Algorithm for Small Constant-depth Circuits with PTF gates
ITCS 2019.

SERVICE AND ENGAGEMENT

Sub-reviewer

SAT 2026.

Sub-reviewer

Computational Complexity Conference (CCC) 2024.

Sub-reviewer

Foundations of Software Technology and Theoretical Computer Science
(FSTTCS) 2024

Volunteer

FSTTCS 2019

TEACHING ASSISTANCE

- Awarded best TA of the semester for CS 101.
- Awarded best TA of the month for Automata Theory.
- TA duty for complexity theory twice, design and analysis of algorithms, combinatorics, and theoretical machine learning.

PROFESSIONAL EXPERIENCE

- Freelance work in quantitative trading for multiple clients since May '22.
- Quantitative Research Consultant for AlphaGrep, and for 5-Swans, during official break from PhD June '21-May '22.
- As Quantitative Developer in AlphaGrep from June '16-July 17 and Jan '14-April '15.
- As Data Scientist in Housing.com from May '15-May '16.
- As Software Engineer in Samsung from July '13-Dec '13.

PERSONAL DETAILS

D.O.B. 09th December 1993

Sex Male

Nationality Indian